In [2]:
```
import pandas as pd
import matplotlib.pyplot as plt

# 1. Read dataset
df = pd.read_csv("City_Air_Quality.csv")

# 2. Explore dataset
print(df.head())
print(df.info())
print(df.describe())

# Convert Date to datetime
df["Date"] = pd.to_datetime(df["Date"])

# 3 & 4. AQI trend
plt.plot(df["Date"], df["AQI"], label="AQI")
plt.xlabel("Date")
plt.ylabel("AQI")
plt.title("AQI Trend Over Time")
plt.legend()
plt.xticks(rotation=45)
plt.show()

# 5. Pollutant trends
plt.plot(df["Date"], df["PM2.5"], label="PM2.5")
plt.plot(df["Date"], df["PM10"], label="PM10")
plt.plot(df["Date"], df["CO"], label="CO")
plt.xlabel("Date")
plt.ylabel("Pollutant Level")
plt.title("Pollutant Trends")
plt.legend()
plt.xticks(rotation=45)
plt.show()

# 6. Bar plot of AQI
plt.bar(df["Date"], df["AQI"])
plt.xlabel("Date")
plt.ylabel("AQI")
plt.title("AQI by Date")
plt.xticks(rotation=45)
plt.show()

# 7. Box plot
df["AQI"] = pd.to_numeric(df["AQI"], errors="coerce")

plt.boxplot(df["AQI"].dropna())
plt.ylabel("AQI")
plt.title("AQI Distribution")
plt.show()

# 8. Scatter plot
plt.scatter(df["PM2.5"], df["AQI"])
plt.xlabel("PM2.5")
plt.ylabel("AQI")
plt.title("PM2.5 vs AQI")
plt.show()
```

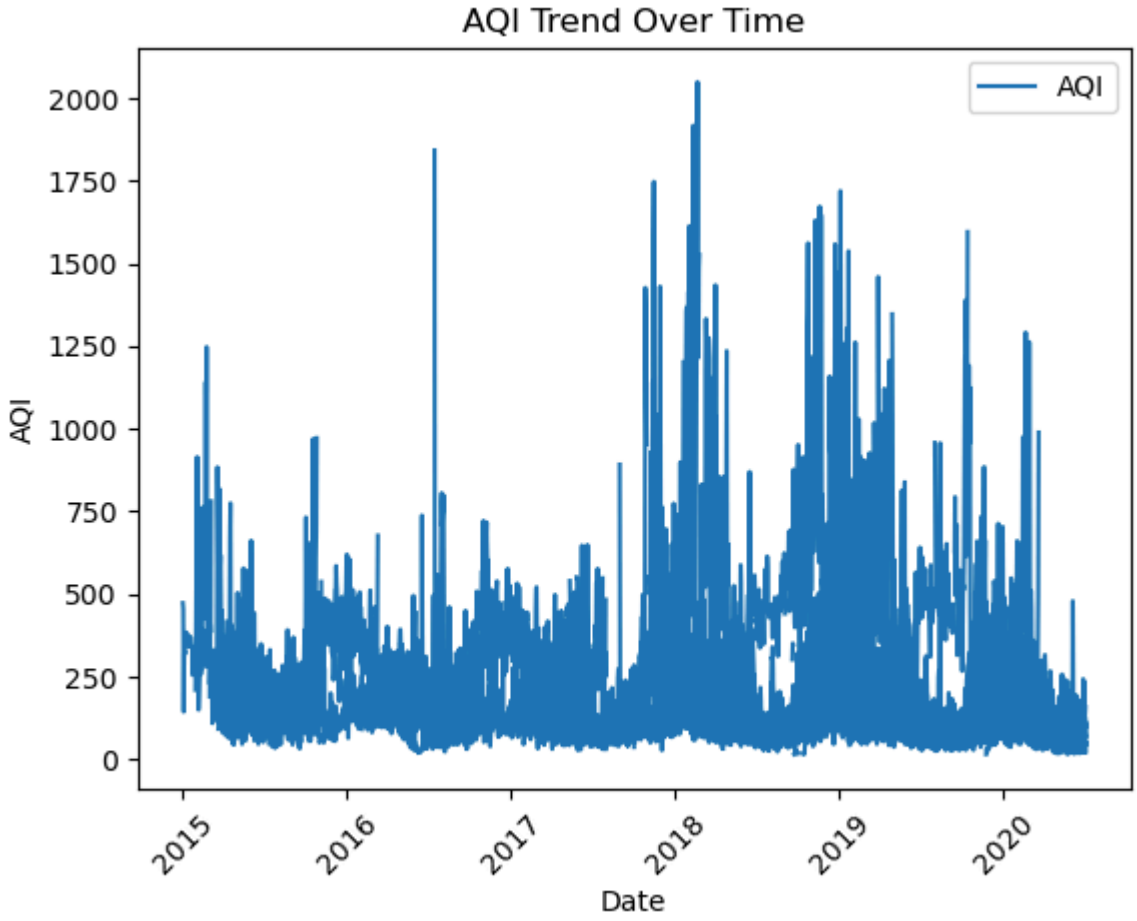
```
        City        Date  PM2.5  PM10     NO    NO2    NOx  NH3     CO    SO2  \
0  Ahmedabad  2015-01-01    NaN   NaN   0.92  18.22  17.15  NaN   0.92  27.64
1  Ahmedabad  2015-01-02    NaN   NaN   0.97  15.69  16.46  NaN   0.97  24.55
2  Ahmedabad  2015-01-03    NaN   NaN  17.40  19.30  29.70  NaN  17.40  29.07
3  Ahmedabad  2015-01-04    NaN   NaN   1.70  18.48  17.97  NaN   1.70  18.59
4  Ahmedabad  2015-01-05    NaN   NaN  22.10  21.42  37.76  NaN  22.10  39.33

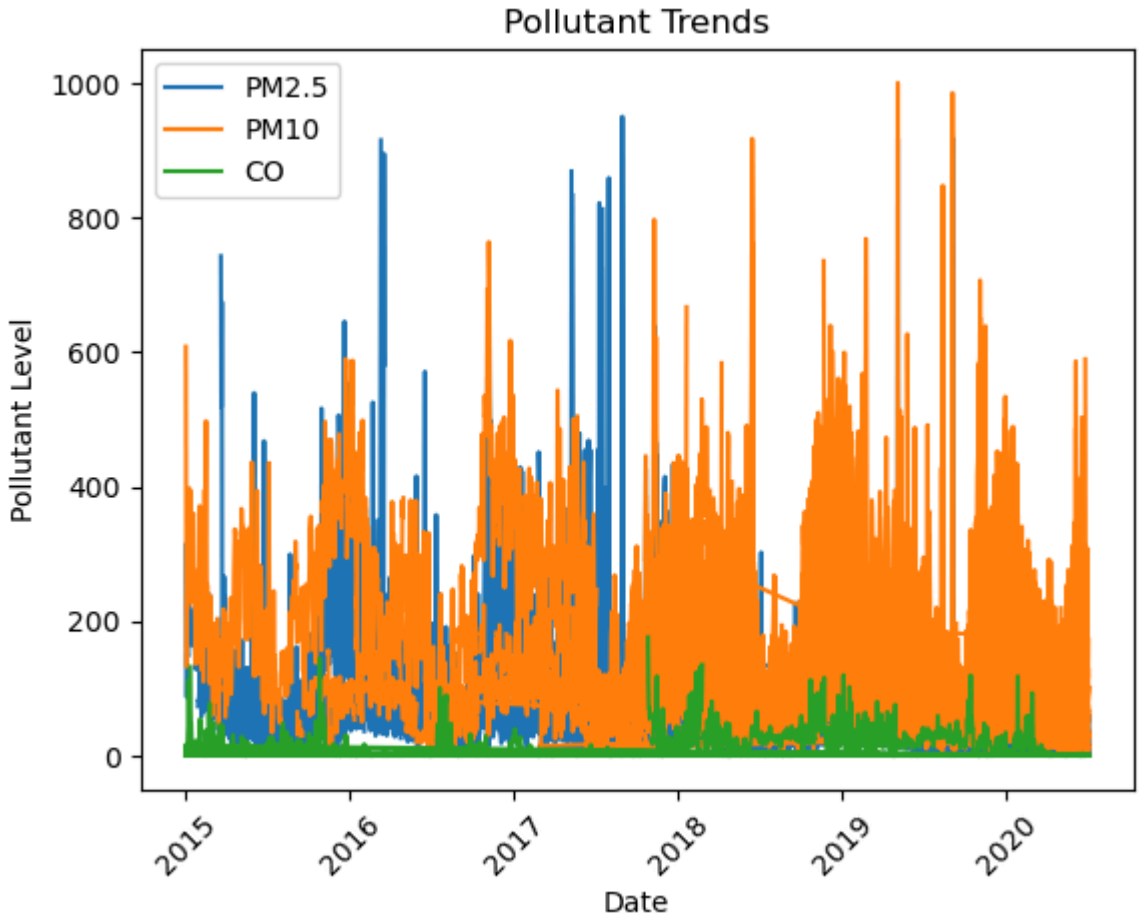
       O3  Benzene  Toluene  Xylene  AQI AQI_Bucket
0  133.36     0.00     0.02    0.00  NaN        NaN
1   34.06     3.68     5.50    3.77  NaN        NaN
2   30.70     6.80    16.40    2.25  NaN        NaN
3   36.08     4.43    10.14    1.00  NaN        NaN
4   39.31     7.01    18.89    2.78  NaN        NaN
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 29531 entries, 0 to 29530
Data columns (total 16 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   City        29531 non-null  object
 1   Date        29531 non-null  object
 2   PM2.5       24933 non-null  float64
 3   PM10        18391 non-null  float64
 4   NO          25949 non-null  float64
 5   NO2         25946 non-null  float64
 6   NOx         25346 non-null  float64
 7   NH3         19203 non-null  float64
 8   CO          27472 non-null  float64
 9   SO2         25677 non-null  float64
 10  O3          25509 non-null  float64
 11  Benzene     23908 non-null  float64
 12  Toluene     21490 non-null  float64
 13  Xylene      11422 non-null  float64
 14  AQI         24850 non-null  float64
 15  AQI_Bucket  24850 non-null  object
dtypes: float64(13), object(3)
memory usage: 3.6+ MB
None
              PM2.5          PM10            NO           NO2           NOx  \
count  24933.000000  18391.000000  25949.000000  25946.000000  25346.000000
mean      67.450578    118.127103     17.574730     28.560659     32.309123
std       64.661449     90.605110     22.785846     24.474746     31.646011
min        0.040000      0.010000      0.020000      0.010000      0.000000
25%       28.820000     56.255000      5.630000     11.750000     12.820000
50%       48.570000     95.680000      9.890000     21.690000     23.520000
75%       80.590000    149.745000     19.950000     37.620000     40.127500
max      949.990000   1000.000000    390.680000    362.210000    467.630000

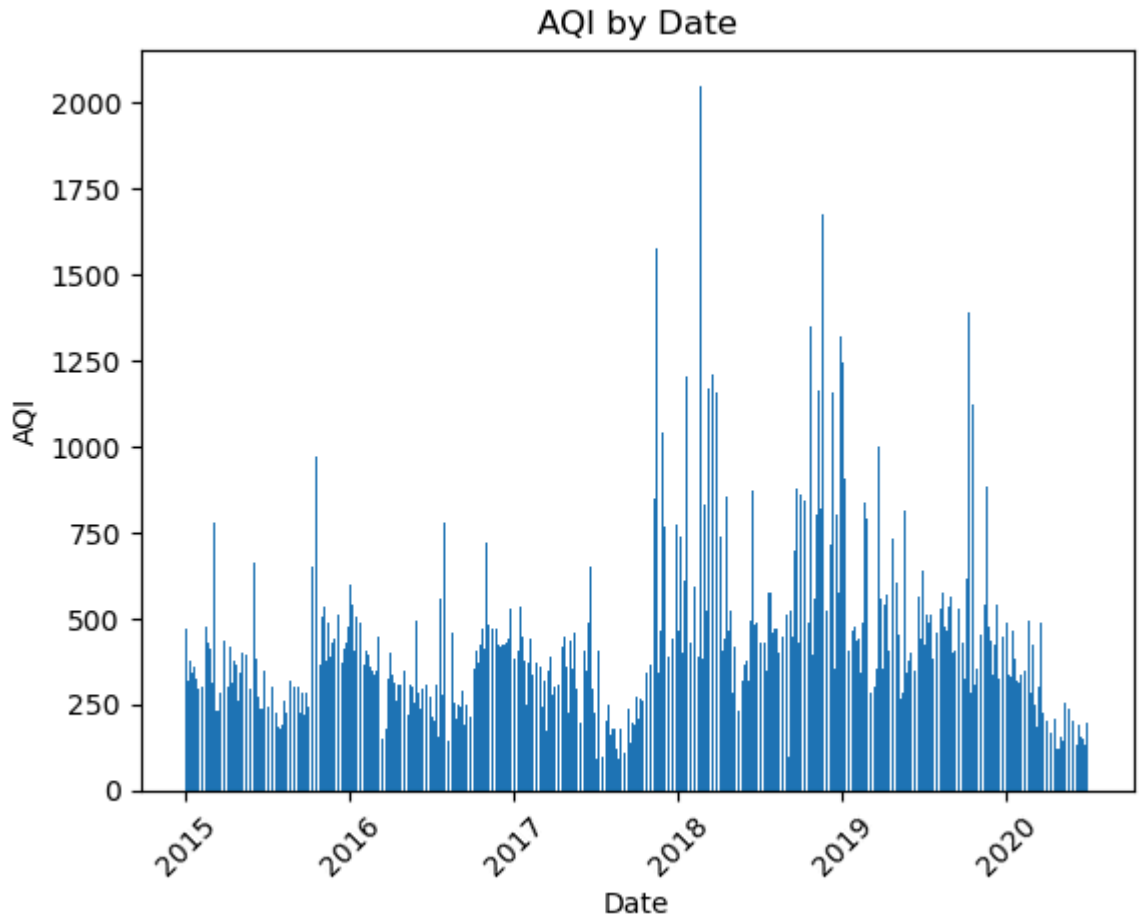
                NH3            CO           SO2            O3       Benzene  \
count  19203.000000  27472.000000  25677.000000  25509.000000  23908.000000
mean      23.483476      2.248598     14.531977     34.491430      3.280840
std       25.684275      6.962884     18.133775     21.694928     15.811136
min        0.010000      0.000000      0.010000      0.010000      0.000000
25%        8.580000      0.510000      5.670000     18.860000      0.120000
50%       15.850000      0.890000      9.160000     30.840000      1.070000
75%       30.020000      1.450000     15.220000     45.570000      3.080000
max      352.890000    175.810000    193.860000    257.730000    455.030000

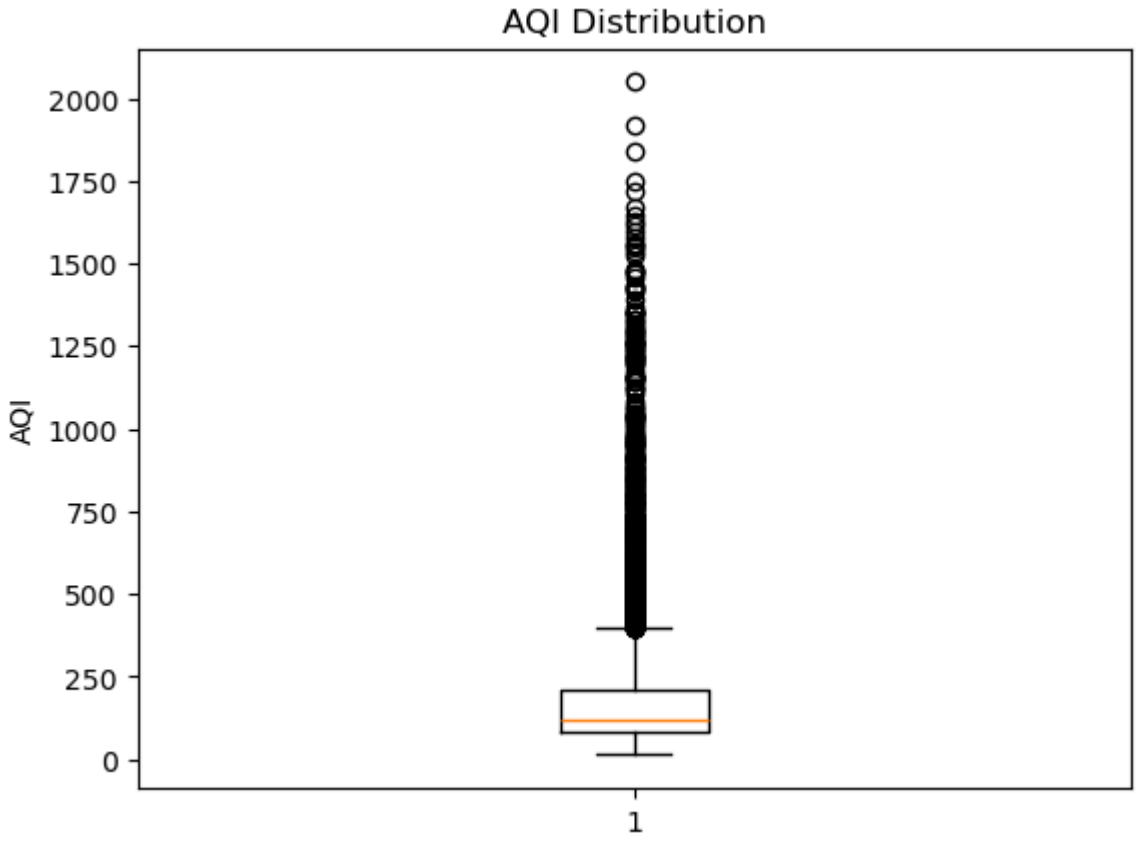
            Toluene        Xylene           AQI
count  21490.000000  11422.000000  24850.000000
mean       8.700972      3.070128    166.463581
```

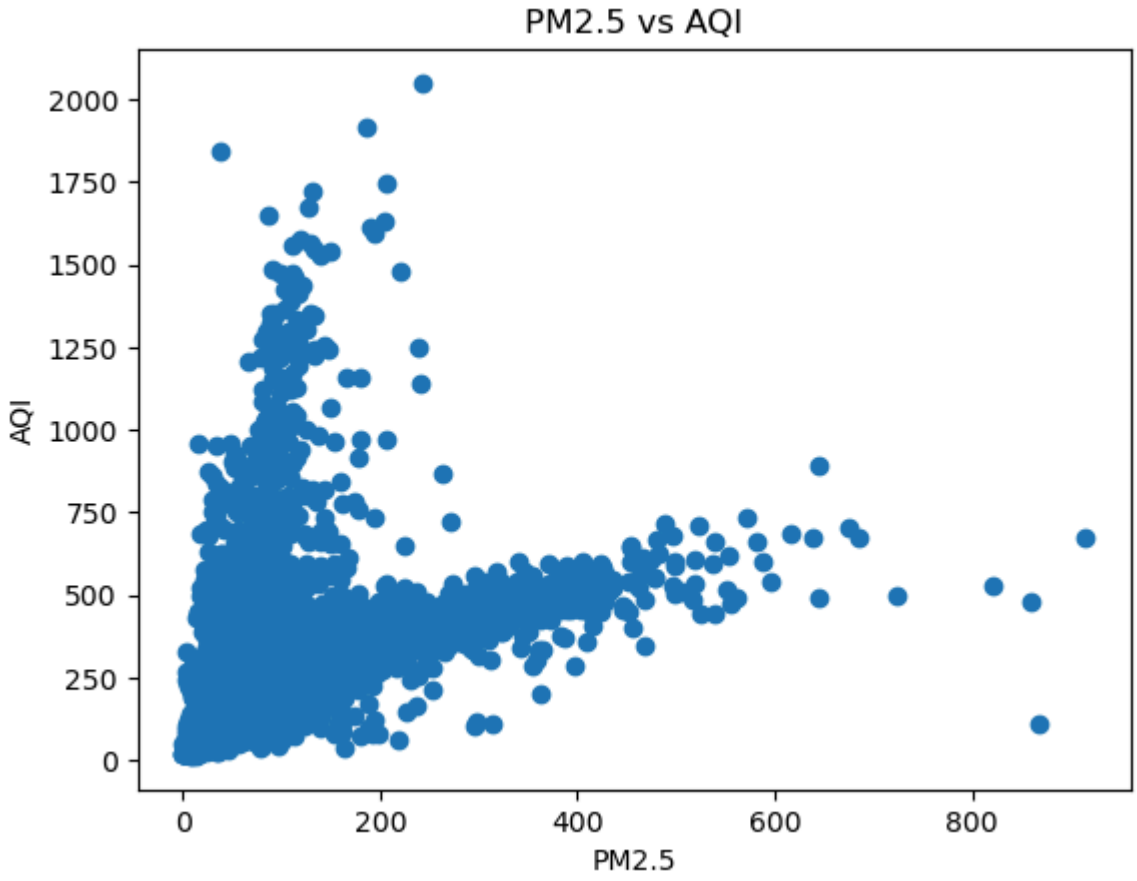
std	19.969164	6.323247	140.696585
min	0.000000	0.000000	13.000000
25%	0.600000	0.140000	81.000000
50%	2.970000	0.980000	118.000000
75%	9.150000	3.350000	208.000000
max	454.850000	170.370000	2049.000000




Pollutant Trends
PM2.5
PM10
CO
Pollutant Level
0
200
400
600
800
1000
2015
2016
2017
2018
2019
2020
Date


AQI by Date
AQI
0
250
500
750
1000
1250
1500
1750
2000
2015
2016
2017
2018
2019
2020
Date


AQI Distribution
AQI
2000
1750
1500
1250
1000
750
500
250
0
1


PM2.5 vs AQI
AQI
2000
1750
1500
1250
1000
750
500
250
0
0
200
400
600
800
PM2.5